Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 7 [2024 P1]

Test Date: 27th Apr 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCED-2024-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Matching Type	+3	-1	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Matching Type	+3	-1	4	12
				17	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Matching Type	+3	-1	4	12
Total				**17**	**60**

MATHEMATICS

Max Marks: 60

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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1. If there are only two points on the hyperbola $2x^2 - 2y^2 - 4x + 41 = 0$, such that tangents drawn to ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a, b \in N)$ are perpendicular, then $|a - b|$ is equal to

A) 2 **B)** 4 **C)** 6 **D)** none of these

2. Number of different terms in the expression $(1+x)^{300} + \left(1+x^2\right)^{200} + \left(1+x^3\right)^{150}$ are

A) 385 **B)** 386 **C)** 384 **D)** 387

3. $\int_{2\pi/3}^{\pi} \sqrt{(\sin x + \sin 2x + \sin 3x)^2 + (\cos x + \cos 2x + \cos 3x)^2}\, dx$ equals:

A) $\left(\sqrt{3} - \frac{\pi}{3}\right)$ **B)** $\left(\frac{\pi}{3} + \sqrt{3}\right)$ **C)** $\frac{5\pi}{3} - \sqrt{3}$ **D)** $\sqrt{3} + \frac{5\pi}{3}$

4. There are four, six faced dice marked 1 to 6 of which two are fair and two are loaded such that the probability of getting a multiple of 3 is twice as probable to get as any other number. A die is chosen at random and rolled. It shows six, the probability that it was biased, is

A) $\frac{1}{8}$ **B)** $\frac{1}{4}$ **C)** $\frac{2}{5}$ **D)** $\frac{3}{5}$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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5. If three planes $P_1 = 2x + y + z - 1 = 0, P_2 = x - y + z - 2 = 0,\ P_3 = 4x - y + \alpha z - 5 = 0$.

Intersect each other at point P on OXY plane at point Q on YOZ plane, where O is origin. Then identity the correct statements

A) the straight line perpendicular to plane P_3 and passing through P is $\frac{x-1}{4} = \frac{y+1}{-1} = \frac{z}{3}$

B) length of projection of $\overrightarrow{PQ}$ on x-axis is 1

C) centroid of ΔOPQ is $\left(\frac{1}{3},-\frac{1}{2},\frac{1}{2}\right)$

D) the distance of the point Q from the line $\frac{x-\frac{47}{6}}{6}=\frac{y-\frac{3}{2}}{2}=\frac{z+\frac{1}{2}}{-4}$ is equal to $\frac{\sqrt{301}}{8}$

6. If $\vec{a}$ and $\vec{b}$ are vectors such that $\left|\vec{a}+\vec{b}\right|=\sqrt{29}$ and $\vec{a}\times\left(2\hat{i}+3\hat{j}+4\hat{k}\right)=\left(2\hat{i}+3\hat{j}+4\hat{k}\right)\times\hat{b}$, then possible value of $\left(\vec{a}+\vec{b}\right)\cdot\left(-7\hat{i}+2\hat{j}+3\hat{k}\right)$ is

A) -3 **B)** 3 **C)** 4 **D)** -4

7. If vector $\vec{a},\vec{b},\vec{c}$ are linearly independent and

$$\left(\vec{d}\times\vec{b}\right)\times\left(\vec{c}\times\vec{a}\right)+\left(\vec{d}\times\vec{c}\right)\times\left(\vec{a}\times\vec{b}\right)+\left(\vec{d}\times\vec{a}\right)\times\left(\vec{b}\times\vec{c}\right)=\vec{r}$$

A) $\vec{r}$ is collinear with $\vec{d}$ **B)** $\vec{d}\cdot\left(\vec{a}\times\vec{r}\right)=0$

C) $\vec{d}\cdot\left(\vec{a}\times\vec{r}\right)\neq 0$ **D)** $\vec{r}$ is a linear combination of $\vec{a},\vec{b}$ and $\vec{c}$

SECTION-III
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

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8. A is a 2×2 matrix such that $A\begin{bmatrix}1\\-1\end{bmatrix}=\begin{bmatrix}-1\\2\end{bmatrix}$ and $A^2\begin{bmatrix}1\\-1\end{bmatrix}=\begin{bmatrix}1\\0\end{bmatrix}$. The sum of the elements of A is

9. Let $f(x)=2\cot x+(2a+1)\ln\left|\operatorname{cosec}x\right|+(2-a)x$ is strictly decreasing in $\left(0,\frac{\pi}{2}\right)$. If range of ' a ' is $[m,\infty)$, then the value of $|m|+1$ is _____

10. If $\int\frac{(x-1)dx}{\left(x+x\sqrt{x}+\sqrt{x}\right)\sqrt{\sqrt{x}(x+1)}}=4\tan^{-1}\left[g(x)\right]+C$, where C is an arbitrary constant of integration. Find $g^2(1)$.

11. If $\left|\vec{a}\right|=1,\left|\vec{b}\right|=3$ and $\left|\vec{c}\right|=4$, then maximum value of $|\vec{a}-\vec{b}|^2+|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2$ is……

12. Let $f(x)$ be a polynomial of degree 5 with integral coefficient that has atleast one integral root. If $f(2)=13$ and $f(10)=5$, then value of α such that $f(\alpha)=0$ (where $\alpha \in I$)

13. The area enclosed by the curve max $(|x-1|,|y|)=4$ is equal to……..

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

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14. Column-I with Column-II (where [.] denotes GIF and {.} denotes fractional part function)

	COLUMN-I		**COLUMN-II**
I)	For $f(x)=\begin{cases}\frac{\lvert x-2\rvert}{x-2}[\log_e x], & x\in[1,3]-\{2\}\\ \{x^2\}, & x\in\left(3,\frac{7}{2}\right]\end{cases}$ number of points of discontinuity is	**P)**	0
II)	For $f(x)=\begin{cases}x^2 & \text{if } x\le 0\\ \sqrt{x} & \text{if } x>0\end{cases}$ number of points of inflexion is	**Q)**	5
III)	Number of points where $f(x)=\max\{\lvert 3-x^2\rvert, 3-x^3\}$ is not differentiable	**R)**	2
IV)	The value of $2\lim\limits_{x\to 0} x^2\left(1+2+3+....+\left[\frac{1}{\lvert x\rvert}\right]\right)$ is equal to where $[x]$ represents GIF	**S)**	1

Codes:

A) I→Q, II→S, III→S, IV→R **B)** I→Q, II→S, III→P, IV→S

C) I→Q, II→S, III)→R,IV→S **D)** I→P, II→Q, III→R, IV→S

15. Match entries of the Column-I with those of Column II

	COLUMN-I		**COLUMN-II**
I)	If slope of focal chord of parabola $y^2=16x$ is 2, then it's length is	**P)**	$\sqrt{2}$
II)	Let PSP' is a focal chord of ellipse $4x^2+9y^2=36$ and $SP=4$ then SP' equals	**Q)**	20

III)	The eccentricity of the hyperbola whose asymptotes are $x+y-1=0$ and $y=x$, is equal to	**R)**	$\sqrt{3}$
IV)	The radius of auxiliary circle of ellipse $x^2+2y^2=3$	**S)**	$\frac{4}{5}$
		T)	2

Codes:

A) I→Q, II→S, III→P, IV→R **B)** I→T, II→Q, III→P, IV→R

C) I→T, II→Q, III)→P,IV→R **D)** I→S, II→T, III→R, IV→P

16. If e_1 and e_2 are the roots of the equation $x^2-ax+2=0$, then match the following:

	COLUMN-I		**COLUMN-II**
I)	If e_1 and e_2 are the eccentricities of the ellipse and hyperbola, respectively, then the values of a can be	**P)**	6
II)	If both e_1 and e_2 are the eccentricities of the hyperbolas, then the values of a can be	**Q)**	$\frac{5}{2}$
III)	If e_1 and e_2 are eccentricities of hyperbola and conjugate hyperbola, then value of $a^2/4$ is	**R)**	$2\sqrt{2}$
IV)	If e_1 is the eccentricity of the hyperbola for which there exists infinite points from which perpendicular tangents can be drawn and e_2 is the eccentricity of the hyperbola in which no such points exist then the values of a can be	**S)**	2

Codes:

A) I→Q, II→S, III→P, IV→R **B)** I→T, II→P, III→Q, IV→S

C) I→P, II→R, III→S, IV→P **D)** I→S, II→T, III→R, IV→P

17. Let P be the plane parallel to y-axis & containing the points (1, 0, 1) & (3, 2, -1). Also $A=(4,0,0)$ & $B=(6,0,-2)$ are 2 pts & $P(x_0, y_0, z_0)$ is a variable point on the plane $P=0$

	COLUMN-I		**COLUMN-II**
I)	If the equation of plane P = 0 is $x+ay+bz=c$ then $\lvert a+b+c\rvert$ is	**P)**	16
II)	If $PA+PB$ is minimum, then $\lvert 4x_0+y_0+2z_0\rvert$ is	**Q)**	12
III)	The minimum value $PA+PB$ is	**R)**	3
IV)	If the reflection of the line AB in the plane P = 0 is $\frac{x-2}{1}=\frac{y-\alpha}{0}=\frac{z-\beta}{-1}$, then $\alpha^4+\beta^4$ is	**S)**	4

Codes:

A) I→R, II→Q, III→S, IV→P **B)** I→Q, II→S, III→R, IV→P

C) I→R, II→P, III→S, IV→Q **D)** I→Q, II→R, III→P, IV→S

PHYSICS **Max Marks: 60**

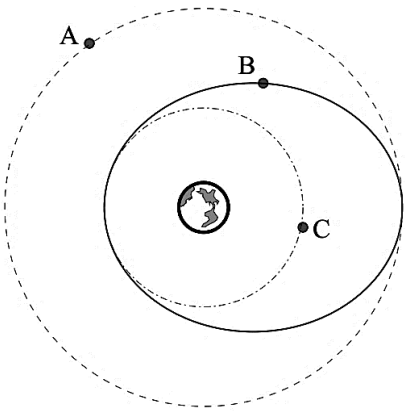
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

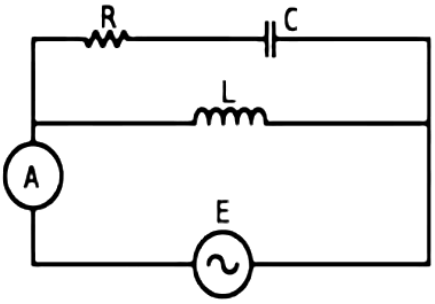
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

18. Three equal mass satellites A, B, and C are in coplanar orbits around a planet as shown in the figure. The magnitudes of the angular momenta of the satellites as measured about the planet are L_A, L_B, and L_C. Which of the following statements is correct?



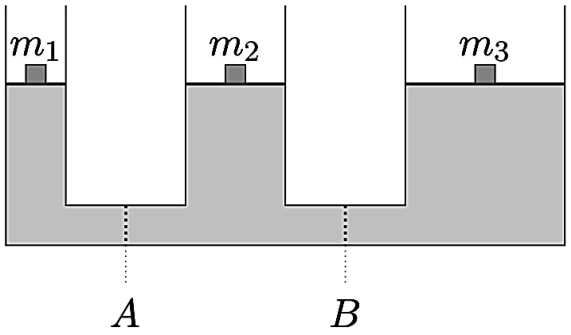
A) $L_A > L_B > L_C$ **B)** $L_C > L_B > L_A$ **C)** $L_B > L_C > L_A$ **D)** $L_B > L_A > L_C$

19. In the circuit shown in the figure the inductance of coil is L = 10 mH and capacitance of the capacitor is C = 0.2 mF. The circuit is powered by an A.C. voltage source. If value of current measured by ammeter does not depend on resistor, then frequency of source is closest:



A) 140 Hz **B)** 130 Hz **C)** 160 Hz **D)** 200 Hz

20. In the device shown, blocks of various masses are placed on pistons so that the device is in equilibrium. (The fluid in the drawing is to scale.) There are valves that are both initially open at locations A and B. One of the valves is closed, and the system is allowed to come to equilibrium. How will this affect the height of the mass m_2?



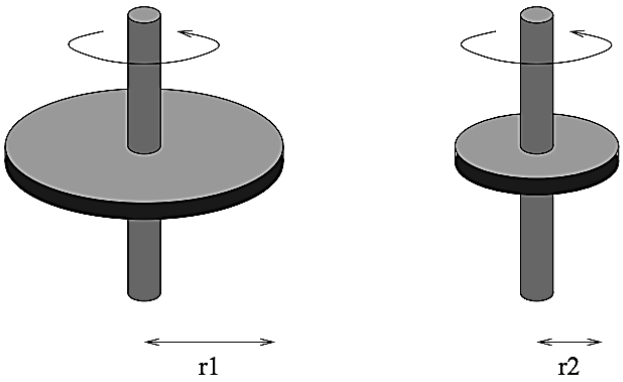
A) Closing either valve will have no effect on the height of m_2

B) Closing either valve causes m_2 to rise.

C) Closing valve A causes m_2 to rise, and closing valve B causes m_2 to fall.

D) Closing valve A causes m_2 to fall, and closing valve B causes m_2 to rise.

21. Two discs are mounted on thin, lightweight rods oriented through their centers and normal to the discs. These axles are constrained to be vertical at all times, and the discs can pivot frictionlessly on the rods. The discs have identical thickness and are made of the same material, but have differing radii r_1 and r_2. The discs are given angular velocities of magnitudes ω_1 and ω_2, respectively, and brought into contact at their edges. After the discs interact via friction it is found that both discs come exactly to a halt. Which of the following must hold? Ignore effects associated with the vertical rods.



A) $\omega_1^2 r_1 = \omega_2^2 r_2$ **B)** $\omega_1 r_1 = \omega_2 r_2$ **C)** $\omega_1 r_1^2 = \omega_2 r_2^2$ **D)** $\omega_1 r_1^3 = \omega_2 r_2^3$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

22. Three blocks of identical mass are placed on a frictionless table as shown. The centre block is at rest, whereas the other two blocks are moving directly towards it at identical speeds *v*. The centre block is initially closer to the left block than the right one. All motion takes place along a single horizontal line.



Suppose that all collisions are instantaneous and may be perfectly elastic or perfectly inelastic. After a long time, which of the following is/are true?

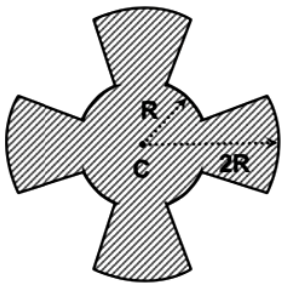
A) If collision is perfectly elastic then the centre block is at rest somewhere to the left of its initial position.

B) If collision is perfectly inelastic then the centre block is at rest somewhere to the right of its initial position.

C) If collision is perfectly elastic then the centre block is at rest at its initial position

D) If collision is perfectly inelastic then the centre block is at rest somewhere to the left of its initial position.

23. A non-conducting portion of circular disc (as shown in figure) having positive charge density σ. Each cross section subtends an angle 45° on its center ' C '. A positive point charge *q* is slowly brought from infinity to the centre ' C '. Choose the correct option(s).



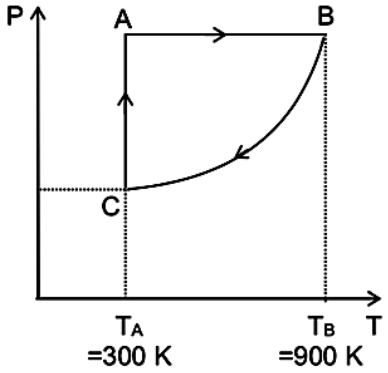
A) The work done by the external agent is positive.

B) The potential due to the non conducting portion at a distance of $\sqrt{3}\mathrm{R}$ on its axis from centre C is $\frac{\sqrt{7}}{4}\frac{\sigma \mathrm{R}}{\varepsilon_0}$.

C) The potential due to the non conducting portion at a distance of $\sqrt{3}\mathrm{R}$ on its axis from centre C is $\frac{\sigma \mathrm{R}}{4\varepsilon_0}\left[2+\sqrt{7}-2\sqrt{3}\right]$.

D) The magnitude of work done by the external agent to bring the charge particle slowly from infinity to the centre is $\frac{3}{4}\frac{\sigma Rq}{\varepsilon_0}$.

24. Two moles of an ideal monoatomic gas is taken through a cyclic process ABCA as shown in the P-T diagram. In the process $BC, P=\alpha T^2$, where α is a constant. Then choose the correct option(s). (R = ideal gas constant) [Take $\ell n(3)=1.1$]



A) The heat absorbed by the gas during the process AB is 3000 R.

B) The heat absorbed by the gas during the process BC is 1200 R.

C) The heat rejected by the gas during the process BC is 600 R.

D) The efficiency of the cycle is 36%.

SECTION-III

(NON-NEGATIVE INTEGER.)

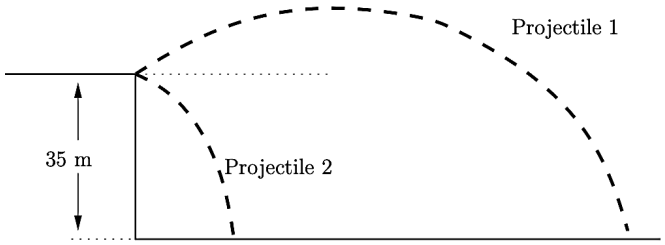
This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

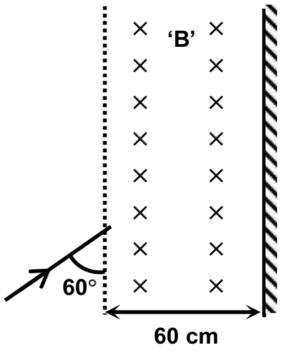
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25. Two projectiles are launched from a 35 meter ledge as shown in the diagram. One is launched from a 37 degree angle above the horizontal and the other is launched from 37 degrees below the horizontal. Both of the launches are given the same initial speed of $v_0 = 50$ m/s.

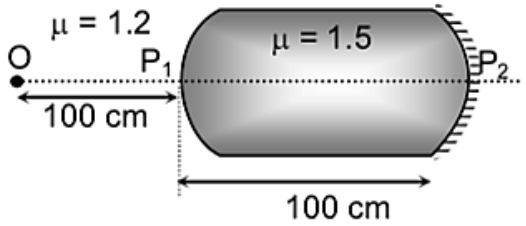


Find the difference in the times (in SI unit) of flight for these two projectiles.
(Take $g = 10\,\text{m/s}^2$)

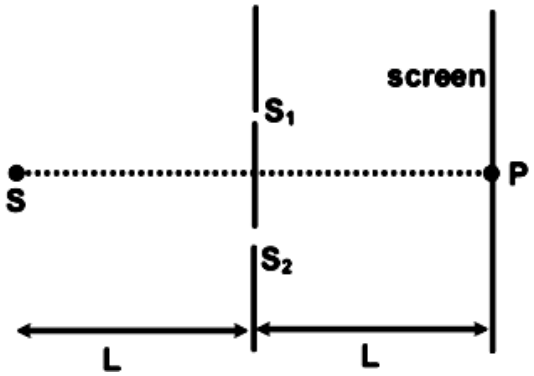
26. A positive charged particle q and mass m enters and comes out from a uniform magnetic field B which exists with direction inside the paper in a finite space of width 60 cm after collision with the wall (collision is perfectly elastic) as shown in the figure. If time elapsed by the charge particle inside the magnetic field is $\left(\frac{23^\circ}{90^\circ}\right)\left(\frac{\pi m}{qB}\right)$. Find the radius of curvature 'R' (in SI unit) of the charge particle when it is inside the magnetic field.



27. A luminous point object O is placed in front of spherical glass surface of radius $R = 20\,\text{cm}$. The final real image is formed after refraction through the spherical surface, followed by reflection at the concave mirror of radius $R = 400\,\text{cm}$ and a second refraction through the spherical surface. Find the distance (in centimetre) of this final image from the pole P_1 of the spherical surface.

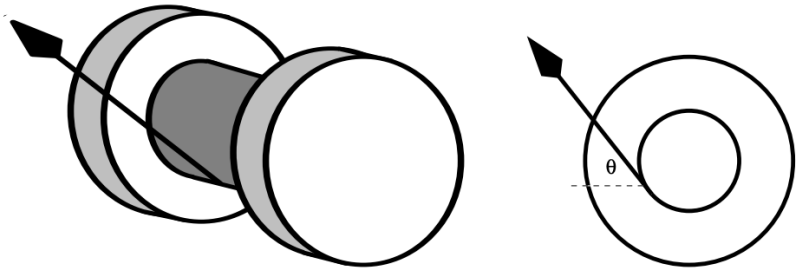


28. Light from a monochromatic point source S is incident on the two slits of an opaque wall placed at a distance L as shown in the diagram. Intensity of interference pattern is observed at the point P on the screen placed at a distance L behind the slits. When the upper slit is shifted upward then the distance between the slits increases. It is observed that when the distance between the slits is changed from $200\,\mu\text{m}$ to $400\,\mu\text{m}$, the intensity at P changes from $I_{\max}$ to $\frac{3}{4}I_{\max}$. If $L = 1\,\text{m}$, then find the wavelength of the light used (in Å)



29. Consider a partially polarized light beam, containing a mix of unpolarized and linearly polarized light. The intensity of the beam is analysed using a linear polarizer. At a particular orientation of the polarizer, the outgoing beam has maximum intensity $I_{\max}$. Turning the polarizer by a 30° angle reduces the outgoing beam's intensity by 10%. The degree of polarization of the beam V is defined as $V \equiv \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$ where $I_{\min}$ is the minimum intensity for any orientation of the polarizer. Fill the value of $20V$

30. The apparatus in the diagram consists of a solid cylinder of radius 1 cm attached at the centre to two disks of radius 2 cm. It is placed on a surface where it can roll, but will not slip. A thread is wound around the central cylinder. When the thread is pulled at the angle $\theta = 90^\circ$ to the horizontal (directly up), the apparatus rolls to the right. Fill the largest value of θ (in degrees) for which it will not roll to the right when pulling on the thread?



SECTION – IV
(MATCHING TYPE)

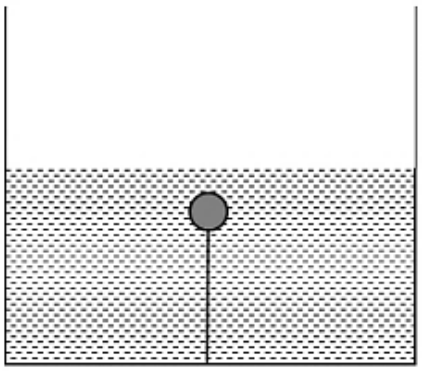
This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

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31. A solid spherical ball is attached to a rigid rod radially and this arrangement is kept inside a liquid (density twice the ball) as shown (with the rod fixed to the container at the bottom): Now the container is given possible accelerations as in List-I. Some numbers are given in List-II.



	COLUMN-I		COLUMN-II
I)	$\vec{a} = \vec{0}$	**P)**	2
II)	$\vec{a} = \frac{g}{2}\downarrow$	**Q)**	1
III)	$\vec{a} = g\downarrow$	**R)**	0
IV)	$\vec{a} = g\uparrow$	**S)**	$\frac{3}{2}$
		T)	$\frac{1}{2}$
		U)	3

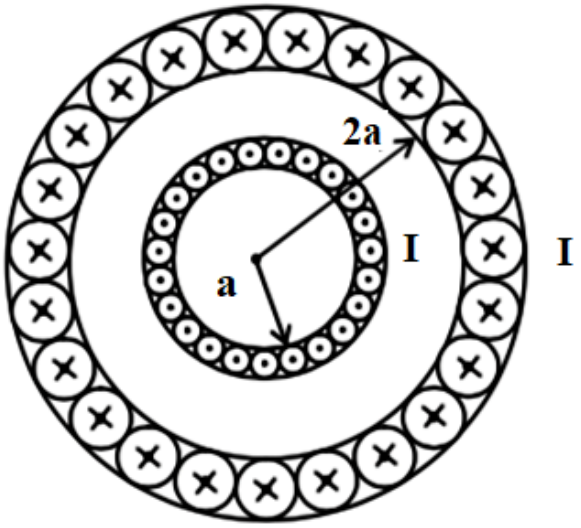
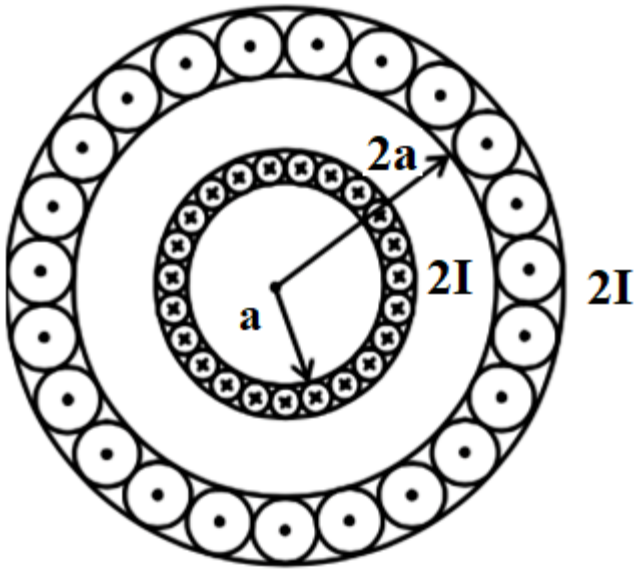
If the experiment is done on earth (acceleration due to gravity: g), then match the ratio $\frac{\text{tension (magnitude) in rod}}{\text{weight (magnitude) of ball on earth}}$. Here, weight of ball on earth means *mass* × *acceleration* due to gravity.

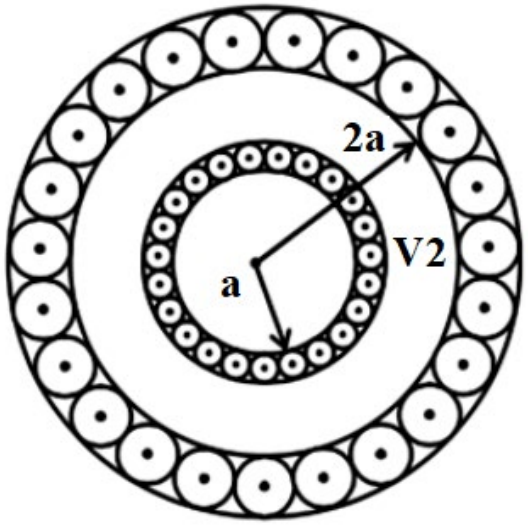
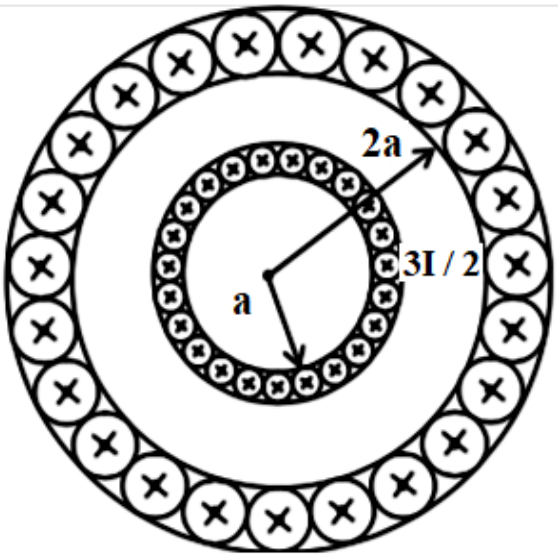
Options

A) I→S, II→Q, III→P, IV→T **B)** I→Q, II→T, III→R, IV→P

C) I→T, II→P, III→R, IV→S **D)** I→T, II→R, III→S, IV→P

32. Two long current carrying thin walled coaxial cylinders are depicted in List-I and List-II gives the magnitude of magnetic pressure exerted on the wall of the cylinders.

	COLUMN-I		**COLUMN-II**
I)		**P)**	$\frac{\mu_0 I^2}{32\pi^2 a^2}$
II)		**Q)**	$\frac{\mu_0 I^2}{8\pi^2 a^2}$

III)		**R)**	$\frac{3\mu_0 I^2}{16\pi^2 a^2}$
IV)		**S)**	$\frac{9\mu_0 I^2}{32\pi^2 a^2}$
		T)	$\frac{\mu_0 I^2}{2\pi^2 a^2}$
		U)	$\frac{9\mu_0 I^2}{16\pi^2 a^2}$

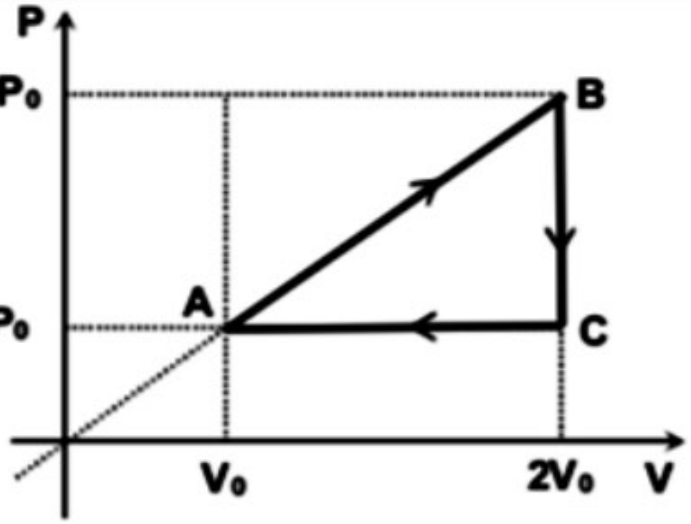
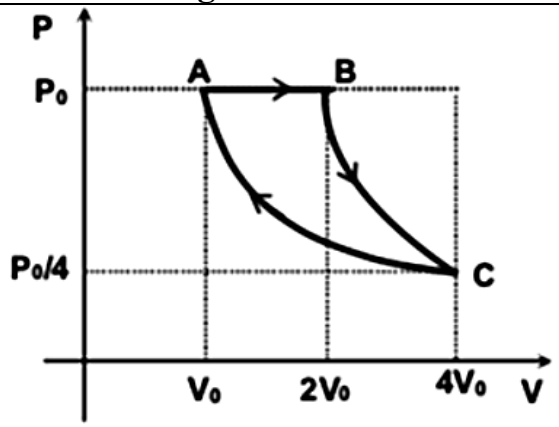
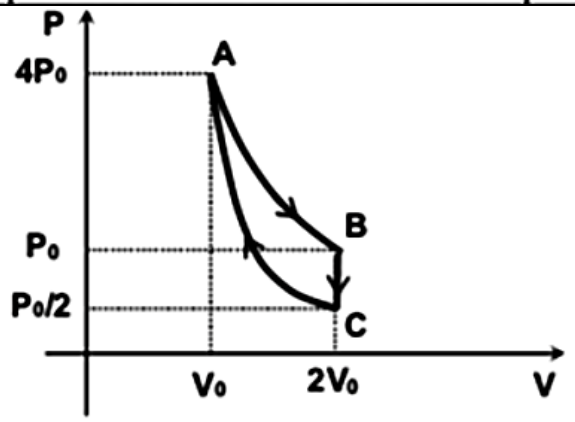
Find the correct match for the magnetic pressure exerted on the wall of each outer cylinder.

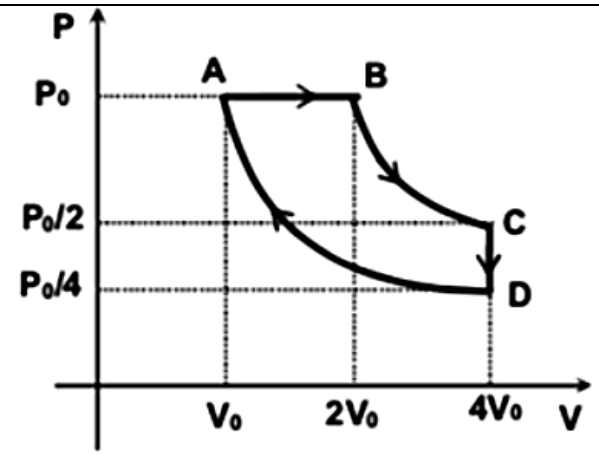
A) I→P, II→Q, III→T, IV→S **B)** I→Q, II→S, III→R, IV→U

C) I→P, II→Q, III)→R,IV→U **D)** I→S, II→T, III→R, IV→P

33. One mole of an ideal gas is taken through four different cyclic processes as depicted in the List-I and List-II gives the values of total work done and total heat absorbed by the gas during four different cyclic processes

	COLUMN-I		**COLUMN-II**

I)	 **monoatomic gas is taken**	**P)**	$\frac{P_0V_0}{2}$
II)	 **monoatomic gas is taken** (For the process BC, PV^2 = constant and process CA is an isothermal process)	**Q)**	P_0V_0
III)	 **Diatomic gas is taken** (For the process AB, PV^2 = constant and for the process CA, PV^3 = constant)	**R)**	$2P_0V_0(1-\ell n2)$

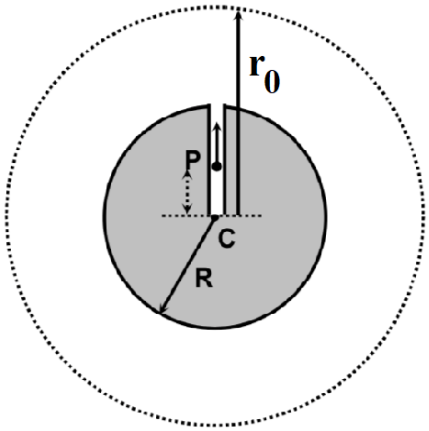
IV)	 **Diatomic gas is taken (Processes BC and DA are isothermal)**	**S)**	$\frac{5P_0V_0}{2}$
		T)	$6P_0V_0$
		U)	$\frac{P_0V_0}{2}(7+4\ell n2)$

Find the correct match for the total work done by the gas during each cyclic process.

A) I→S, II→R, III→P, IV→T **B)** I→P, II→R, III→P, IV→Q

C) I→P, II→Q, III→R, IV→T **D)** I→Q, II→P, III→S, IV→R

34. A satellite is to be launched in a circular orbit of radius r_0 around a planet of radius R. For this two stage rocket is to be used. First stage rocket is used to lift the satellite vertically up to the desired height r_0 (measured from the center C of the planet), after launching it from a launch point P. Distance of launch point P from centre of the planet is r_L. Launch point P lies inside the tunnel as shown in the figure. Second stage rocket is used to impart it a tangential velocity so as to put it in a circular orbit. Let E_1 and E_2 are the energies delivered by the first and second stage of the rocket. Assume that the mass of the rocket is negligible with respect to the mass of the planet and atmospheric resistance to be zero. List-I gives 4 values of r_0 and r_L and List-II gives magnitude of some quantity



	COLUMN-I		COLUMN-II
I)	$r_0 = R \quad r_L = \frac{R}{\sqrt{2}}$	**P)**	$\frac{1}{2}$
II)	$r_0 = \frac{3R}{2} \quad r_L = R$	**Q)**	1

III)	$r_0 = 4R \quad r_L = R$	**R)**	$\frac{11}{6}$
IV)	$r_0 = 4R \quad r_L = \frac{R}{\sqrt{2}}$	**S)**	4
		T)	6
		U)	8

The correct match for $\frac{E_1}{E_2}$ for the values of r_0 and r_L given in List-I is Options

A) I→S, II→T, III→U, IV→P **B)** I→P, II→Q, III→T, IV→U

C) I→S, II→T, III→R, IV→P **D)** I→P, II→S, III→Q, IV→T

CHEMISTRY **Max Marks: 60**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

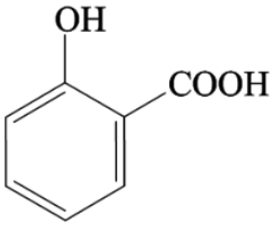
This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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35. Salicylic acid is a weak organic acid which is found inside the bark of willow tree. The structure of salicylic acid is shown below:



Salicylic acid is a precursor to aspirin, a popular NSAID used to combat pain and fever.

In order to make aspirin, salicylic acid (pka = 3) mixed with acetic acid (pka = 4.74)

Choose the most likely pka value of for acetylsalicyclic acid (aspirin):

A) 1.5 **B)** 2.5 **C)** 3.5 **D)** 10.0

36. The standard reduction potentials of some half-cell reactions are given below:

$PbO_2 + 4H^+ + 2e^- \rightleftharpoons Pb^{2+} + 2H_2O$ $\quad E^o = 1.455\,V$

$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O$ $\quad E^o = 1.51\,V$

$Ce^{4+} + e^- \rightleftharpoons Ce^{3+}$ $\quad E^o = 1.61\,V$

$H_2O_2 + 2H^+ + 2e^- \rightleftharpoons 2H_2O$ $\quad E^o = 1.71V$

Pick out the wrong statement:

A) Ce^{+4} will oxidize Pb^{2+} to PbO_2

B) MnO_4^- will oxidize Pb^{2+} to PbO_2

C) H_2O_2 will oxidize Mn^{+2} to MnO_4^-

D) PbO_2 will oxidize Mn^{+2} to MnO_4^-

37. Identify the true statement regarding Indian saltpeter:

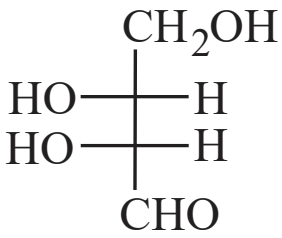
A) it imparts lilac colour to the Bunsen flame

B) it is the minor constituent of gunpowder

C) upon heating it evolves a diamagnetic gas

D) it is obtained naturally in triclinic crystal habit at room temperature

38. Identify the correct statement regarding the following carbohydrate:



A) it has L-configuration

B) it shows negative Tollen's test

C) it is oxidised to meso-tartaric acid with conc. HNO_3

D) it can be made by reacting E-stereoisomer of crotonaldehyde with SeO_2 followed by adding Baeyer's regent

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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39. K_p for the reaction: $NH_4I(s) \rightleftharpoons NH_3(g) + HI(g)$ is $\frac{1}{4}(atm)^2$ at 300 K.

If 4 moles of $NH_4I(s)$ is taken in a 100 litre container, identify the correct conclusion (s):

(assume $R = \frac{1}{12}$ atm lit mole^{-1}K^{-1})

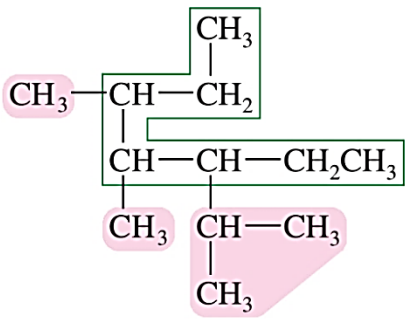
A) the total pressure at equilibrium is 1 atm

B) 2 moles $NH_4I(s)$ undergoes dissociation to gases

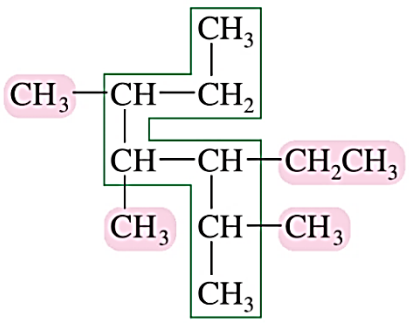
C) all $NH_4I(s)$ undergoes dissociation to gases (no equilibrium)

D) a total 4 moles of hydrogen containing gases are produced

40. In a structural isomer of Dodecane, the longest carbon chain can be selected in two different ways as shown in the marked enclosures below



(I)



(II)

Identify the right conclusion(s) regarding the above alkane

A) it contains an equal number of secondary and tertiary hydrogens

B) it does not contain any quaternary carbon atom

C) (I) is the correct way to select the longest carbon chain for IUPAC nomenclature

D) the shaded entities are the substituents to the proposed main chain

41. Which molecules have the same hybridisation for their central atom as that of ozone (O_3) and contain more lone pair of electrons than present in the overall structure of O_3.

A) BHF_2 **B)** BCl_3 **C)** SO_3 **D)** NI_3

SECTION-III

(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

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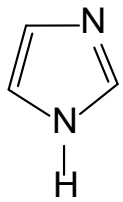
42. For a unidirectional first order decomposition: $X(g) \longrightarrow Y(g) + 2Z(g)$ the following P-t data is plotted. (P = total pressure)

t (min)	0	20	∞
P (mm)	100	220	250

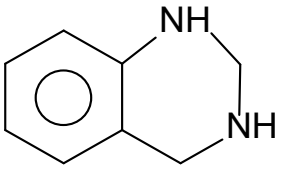
Calculate $t_{\frac{24}{25}}$ (time taken for $\frac{24}{25}$th of the reactant to decompose) in minutes.

43. Each of the following organic bases contain an even number of nitrogen atoms. Select the ones that contain an equal number of localized and delocalized lone pair and report the sum of the serial number of all such molecules.

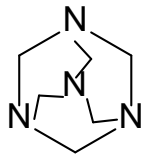
1



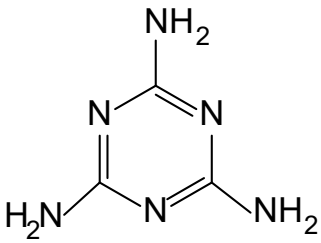
2



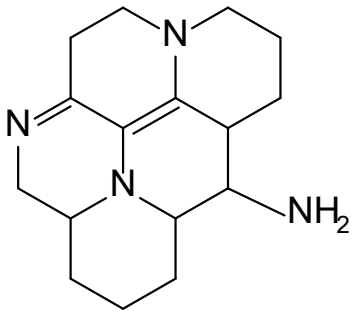
3



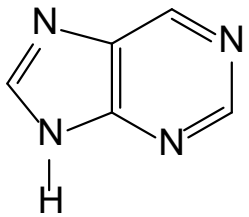
4



5



6



44. Crude calcium carbide is made in an electric furnace as per the following reaction:

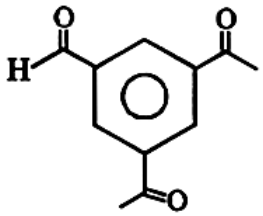
$$CaO + 3C \longrightarrow CaC_2 + CO$$

The crude product contains 64% CaC_2 and 36% unreacted CaO by weight.
How many kilograms of CaO must be added to the furnace for each $1000\,kg$ of the crude product.

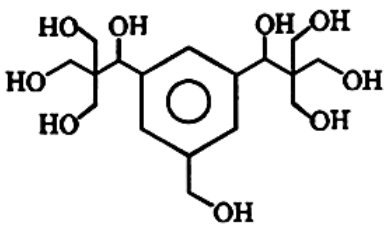
45. An organic compound $X(C_7H_{16}O)$ gives effervescence with Na. X has both enantiomers and diastereomers. X gives yellow precipitate with alkaline iodine solution. With $CrO_3 - H_2SO_4$, X is converted into $C_7H_{14}O(Y)$ which has enantiomers but not diastereomers. X on refluxing with H_2SO_4 isomerises to Z which neither gives yellow precipitate with alkaline iodine nor changes colour of $CrO_3 - H_2SO_4$. What is the minimum number of carbons that can be present in the parent chain of X?

46. pH of a $0.1M$ solution of a monobasic acid is 2.0. Its osmotic pressure at a given temperature T K is XRT then the value of 100 X is _____

47. The number of moles of HCHO required to convert 1 mole of **X** completely into **Y** when X is treated with conc. NaOH/HCHO is



X



Y

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases..

48.

COLUMN - I Colour with $K_4[Fe(CN)_6]$		COLUMN - II Cation	
(I)	white precipitate	**(P)**	Cu^{+2}
(II)	blue precipitate	**(Q)**	Ca^{+2}
(III)	brown precipitate	**(R)**	Fe^{+2}
(IV)	no precipitate	**(S)**	Sr^{+2}
		(T)	Fe^{+3}

Codes:

A) $(I)-Q\ ;(II)-P;(III)-R;(IV)-T$

B) $(I)-Q,\ R\ ;\ (II)-T;(III)-P;(IV)-S$

C) $(I)-Q,\ T;(II)-T\ ;(III)-P;(IV)-R$

D) $(I)-Q;(II)-R;(III)-P;(IV)-T$

49.

Column-I Solid Solution		**Column-II Properties of constituent atoms**	
(I)	Brass	**(P)**	occupy adjacent boxes in the periodic table
(II)	Bronze	**(Q)**	occupy diagonal boxes in the periodic table
(III)	German silver	**(R)**	Sum of atomic numbers is a prime number
(IV)	Silver amalgam	**(S)**	Make a binary solution
		(T)	One of the atoms have exceptional electronic configuration

Codes:

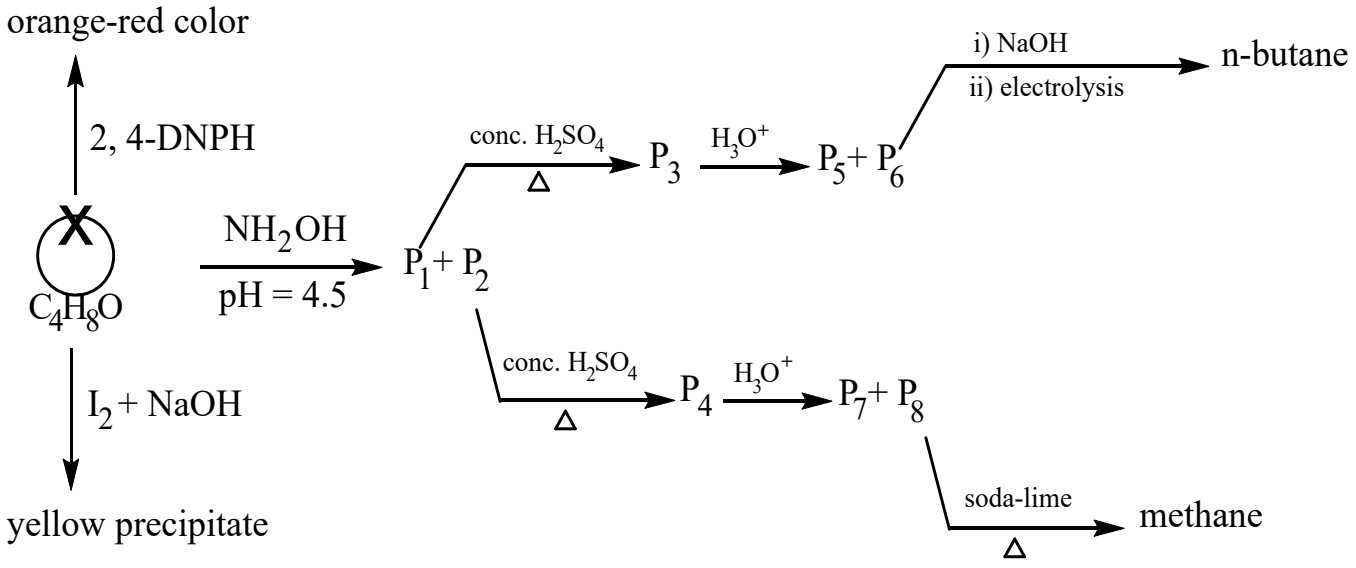
A) $(I)-P,T\ ;\ (II)-R,T\ ;\ (III)-P;(IV)-R,T$

B) $(I)-P,\ R,\ T;\ (II)-R,\ S,\ T;(III)-P,\ R,\ T;(IV)-Q,\ S,T$

C) $(I)-P,\ R,\ S,\ T;(II)-S,\ T;(III)-P,\ T;\ (IV)-Q,\ R,\ S$

D) $(I)-\ P,\ R,\ S,\ T;(II)-R,\ S,\ T;(III)-P,\ T;(IV)-Q,\ R,\ S,T$

50.



	Column-I (Information)		Column-II (Corresponding pair(s) of molecules)
(I)	Pair of structural isomers	**(P)**	P_1 & P_2
(II)	Pair of stereoisomers	**(Q)**	P_3 & P_4
(III)	Pair of adjacent homologues	**(R)**	P_5 & P_7
(IV)	Pair of compounds that show positive carbylamine test	**(S)**	P_6 & P_8
		(T)	P_1 & P_3

Codes:

A) $(I)-Q,\ T;(II)-P;(III)-R,\ S;(IV)-R$

B) $(I)-Q;(II)-P;(III)-R;\ (IV)-S$

C) $(I)-T;(II)-Q;\ (III)-S;\ (IV)-R$

D) $(I)-Q,\ T;\ (II)-P;\ (III)-R,\ S;\ (IV)-S$

51.

	Column-I Chemical Species		Column-II Standard enthalpy & entropy of formation
(I)	$Br_2(\ell)$	**(P)**	$\Delta H_f^\circ = 0\ ;\ \Delta S_f^\circ = 0$
(II)	$O_3(g)$	**(Q)**	$\Delta H_f^\circ > 0\ ;\ \Delta S_f^\circ > 0$
(III	$I_2(g)$	**(R)**	$\Delta H_f^\circ < 0\ ;\ \Delta S_f^\circ < 0$
(IV)	$CO(g)$	**(S)**	$\Delta H_f^\circ > 0\ ;\ \Delta S_f^\circ < 0$
		(T)	$\Delta H_f^\circ < 0\ ;\ \Delta S_f^\circ > 0$

Codes:

A) $(I)-P;\ (II)-S;\ (III)-Q;\ (IV)-T$

B) $(I)-P;\ (II)-S;\ (III)-P;\ (IV)-T$

C) $(I)-P;(II)-P;\ (III)-Q;\ (IV)-T$

D) $(I)-R;\ (II)-S;\ (III)-Q;\ (IV)-S$